INTERNATIONAL MONETARY FUND

GHANA

STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITYANALYSIS

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Prepared by the International Monetary Fund and the World Bank[1]

Based on an assessment of external public debt indicators, Ghana still faces a moderate risk of debt distress, but overall debt vulnerabilities have increased, and Ghana's debt service-to-revenue ratio is approaching high-risk levels. Driven by loose fiscal policy, deteriorating financing terms and external pressures, several of Ghana's public domestic and external debt indicators have deteriorated. Assuming fiscal policies are implemented as planned, total public debt is projected to hover around 55 to 60 percent of GDP, with rising debt service absorbing more than 40 percent of government revenue in the long run. The external debt service-to-revenue ratio temporarily breaches and subsequently stays close to its indicative threshold in the long term. Additional consolidation measures of about 2 percent of GDP in 2014, combined with a more ambitious medium-term adjustment, would greatly reduce the risk of worsening debt and debt service indicators.

[1] Prepared in collaboration with Ghanaian authorities. The previous DSA was prepared in May 2013 (IMF Country Report No. 13/187)

KEY ASSUMPTIONS UNDER THE BASELINE SCENARIO

1. Macroeconomic assumptions are broadly in line with those in the 2013 DSA, with deviations mainly related to the projection of oil production and higher cost of borrowing (Box 1, Tables 1–2). Revisions reflect: (i) updated information on the timing of oil production; (ii) an associated upward revision in real growth for the medium term and lower growth subsequently; and (iii) higher cost of borrowing, consistent with current lending conditions and projected rising world interest rates in the medium term. As projections on future oil discoveries and gas production are not available (and thus, not captured), the growth projections are conservative.

Table 1. Key Macroeconomic Assumptions

	2013	2014	2014-19	2020-33
Real Growth		**(annual percentage change)**		
DSA-2013	7.9	6.1	5.8	5.3
DSA-2014	5.4	4.8	6.1	4.7
Inflation (GDP deflator)		**(annual percentage change)**		
DSA-2013	13.7	12.8	9.6	6.9
DSA-2014	12.3	13.1	10.2	6.8
Real interest Rate (foreign debt)		**(percent)**		
DSA-2013	1.6	1.1	1.7	3.7
DSA-2014	3.0	2.4	2.6	3.8
Current account balance		**(in percent of GDP)**		
DSA-2013	-11.9	-9.1	-8.3	-6.0
DSA-2014	-13.2	-10.6	-7.7	-6.0
Primary fiscal balance		**(in percent of GDP)**		
DSA-2013	-6.5	-5.1	-2.6	-0.4
DSA-2014	-5.6	-3.6	-0.9	-0.3

2. The projected debt dynamics are, however, contingent on successful fiscal consolidation. The baseline assumes fiscal consolidation that is consistent with the authorities' 2014 nominal budget targets and stated policies, but a more conservative growth outlook and higher interest payments. This implies a gradual reduction in the fiscal deficit to 7½ percent of GDP in 2016, with subsequent changes driven primarily by the profile for oil revenue.

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

Baseline scenario

3. Ghana's external debt indicators are broadly in line with the 2013 DSA, with PV indicators moderated by a change in the discount rate (Table 3 and Figure A1). The deterioration in the PV of debt stock indicators compared to last year's assessment was mitigated by an increase in the discount rate

Box 1. Baseline Macroeconomic Assumptions

Real GDP-growth: While still solid, real growth has declined to a projected 5½ percent in 2013, due to disruptions in power supply from the West African Gas Pipeline in the first half of the year, crowding out from high real interest rates, higher import costs due to the depreciation, and falling gold production related to the drop in world prices. In 2014-15, growth is projected to remain at this more moderate level until production from new oil fields boosts output in 2016. In the long run, real growth is assumed to stabilize at 4¾ percent, with new oil discoveries and the associated gas production implying significant upside potential.

Inflation: Inflation overshot the end of the year target range of 9+/-2 percent (end-year CPI inflation was 13.5 percent, partly due to large increases in utility prices), and is projected to remain at this elevated level in 2014. Supported by an improvement in the policy mix, inflation is expected to decline in the medium run, with the GDP deflator stabilizing at 6¾ percent in 2020–2033.

Government balances: The overall fiscal balance fell only slightly to an estimated 10.9 percent of GDP in 2013, with interest expenditures amounting to some 5 percent of GDP. Based on the government's policy pronouncements and updated macro-projections, including lower growth and higher interest rates than assumed in the budget, the deficit is expected to stay above 10 percent of GDP in 2014 (primary deficit of 3½ percent). Increased revenues from the production of oil and further consolidation in 2015 and 2016 would balance primary expenditures and revenues in the medium to long term.

Current account balance: On the back of a higher fiscal deficit, declining gold and cocoa prices and large oil imports due to electricity disruptions in the first half of the year, the 2013 current account deficit rose above 13 percent of GDP. Owing to weak terms of trade, the deficit is expected to remain above 10 percent of GDP in 2014 and elevated until 2015. With increased oil production, the deficit is projected to decline to about 7 percent of GDP by 2017 and to fall gradually to about 4 percent of GDP in the long run, broadly in line with its optimal level according to staff's external balance assessment. Reserves would build up to about 4 months of imports in the long run, consistent with the government's own medium-term target.

Financing flows: While still significant, FDI is estimated to have declined to about 7¼ percent of GDP in 2013. Driven by the discovery of new fields, FDI is projected to stay close to 7½ percent of GDP until 2018, and then to gradually decline to 4 percent of GDP in the long run. Consistent with Ghana's improving income status and wealth, inflows from grants are projected to decline to less than ½ percent of GDP in the medium to long term. Borrowing is projected to become increasingly nonconcessional, with rates for external commercial borrowing revised upwards by 1 percentage point compared to the last DSA.

Table 2. Official Borrowing Assumptions (in Millions of U.S. Dollars)

	2014	2015	2016	2017	2018	2019
Disbursements						
Multilateral	365	518	596	380	361	343
Bilateral	231	474	545	339	339	339
Commercial	1131	963	1403	2178	1822	1532
of which: CDB	*525*	*567*	*567*	*395*	*400*	*0*
Total	1955	2543	2897	2522	2214	2371

from 3 to 5 percent.[2] Export related indicators improve related to the update in the oil production profile. Most importantly, the debt service-to-revenue indicator deteriorates compared to the last assessment due to expected rising costs of external commercial borrowing (1 percentage point above the assumptions in the last DSA), the issuance of the 2013 Eurobond, and the weakening of the cedi.

Table 3. Indicators of External Debt Vulnerabilities (Baseline)

	2018	2024	2033
PV of debt-to-GDP ratio			
DSA-2013	26	29	24
DSA-2014	32	29	23
Threshold	*50*	*50*	*50*
PV of debt-to-exports ratio			
DSA-2013	83	96	76
DSA-2014	70	71	56
Threshold	*200*	*200*	*200*
PV of debt-to-revenue ratio			
DSA-2013	121	137	110
DSA-2014	134	134	112
Threshold	*300*	*300*	*300*
Debt service-to-exports ratio			
DSA-2013	6	14	14
DSA-2014	7	11	11
Threshold	*25*	*25*	*25*
Debt service-to-revenue ratio			
DSA-2013	9	19	20
DSA-2014	13	20	22
Threshold	*22*	*22*	*22*

4. The external debt service-to-revenue ratio breaches its indicative threshold temporarily in the long term, but a probability approach confirms that the breach is not significant (Figure A2). Without assuming measures to smooth the amortization of the 2013 Eurobond, the 2023 bullet repayment results in a breach in the indicative external debt service-to-revenue ratio. Issuance of another Eurobond in 2014 would lead to a breach in two consecutive years in the absence of measures to smooth the amortization profile, and to a possibly more sustained breach of the same threshold towards the end of the projection period. A complementary probability approach, which assesses Ghana's external debt sustainability based on an indicative threshold derived from Ghana's own institutional and growth characteristics, suggests that this breach is minor and temporary (1 year). Thus, the breach, which could be smoothed out in the course of the next 10 years, does not warrant a change in Ghana's assessed risk of debt distress, but is nonetheless indicative of the longer-term perils of continued resorting to market financing to finance recurrent fiscal deficits.

Standard stress tests

5. Standard stress tests confirm a moderate risk of debt distress (Figure A1 and A2, Table 3A). All three stock indicators as well as the external debt service-to-export ratio remain under their respective thresholds even under the standardized stress tests. However, the external debt service to-revenue-ratio—temporarily breaching its threshold level in the baseline scenario—increases to above 30 percent under the most extreme shock which constitutes a one-time 30 percent real depreciation relative to the baseline in 2015. While having a more moderate impact, standardized shocks to non-debt creating flows, growth, or financing conditions would also imply breaches of the debt service-to-revenue threshold.

[2] The discount rate was increased to 5 percent as of October 2013.

B. Public Debt Sustainability Analysis

Baseline scenario

6. Additional risks arise from the high level and rising trend of domestic debt, as indicated by the total public debt dynamics (Table 4, Figure A3). Despite the upward revision in the discount rate to 5 percent, stock indicators of public debt are still close to the 2013 assessment levels. However, owing to deteriorated domestic and external borrowing conditions, and in line with the external debt sustainability assessment, the total public debt service-to-revenue ratio (including payments on external and domestic debt) has deteriorated. It is now projected to increase to 45 percent in the long run, thus absorbing a large part of revenues and leaving the country vulnerable to shocks.

Table 4. Indicators of Public Debt Vulnerabilities (Baseline)

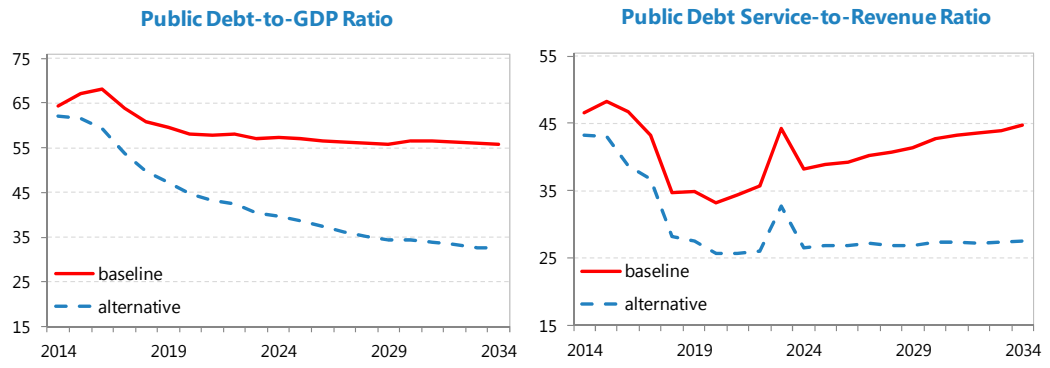
	2018	2024	2033
PV of debt-to-GDP ratio			
DSA-2013	53	56	52
DSA-2014	56	54	54
Benchmark	*74*	*74*	*74*
PV of debt-to-revenue ratio			
DSA-2013	240	257	243
DSA-2014	235	248	270
Debt service to revenue ratio			
DSA-2013	29	38	40
DSA-2014	35	38	45

Stress tests and customized scenario

7. Fiscal policy will have to be tighter than in the past to prevent unsustainable debt dynamics (Figure A3, Table A4). Fixing the primary balance at its 2013 level, or growth and the primary balance at 10-year historical averages, confirms that the historical fiscal stance would not be sustainable in the long run; the PV of public debt-to-GDP ratio would be quickly approaching the benchmark (set at 74 percent for strong policy performers), and debt service would soon absorb most of revenues. Consistent with the results of the external debt sustainability analysis, a one-time 30 percent real depreciation in 2015, relative to the baseline, is the most extreme shock. It would set the debt service-to-revenue ratio on a rapidly increasing path, in the absence of corrective policy measures. The standardized shock to other debt creating flows captures contingent liabilities that could arise from SOE or other potential claims that have not been identified. Any potential financial sector contingent liabilities would also be subsumed in such a standardized shock. This shock would shift debt ratios upwards, but not change the dynamics.

8. Further consolidation of about 2 percentage points of GDP in 2014, combined with a more ambitious medium-term adjustment, would set the debt dynamics on a more favorable path (Figure 1). The additional fiscal adjustment would set off a virtuous cycle where lower fiscal deficits and falling interest rates would create room for higher social and infrastructure spending, a reduction in tax rates, and a crowding in of private sector activity. As a result, growth would be higher, the public debt ratio would decline, and debt service would fall below 30 percent of revenue in the long run.

Figure 1. Public Debt Vulnerabilities (Active Scenario)



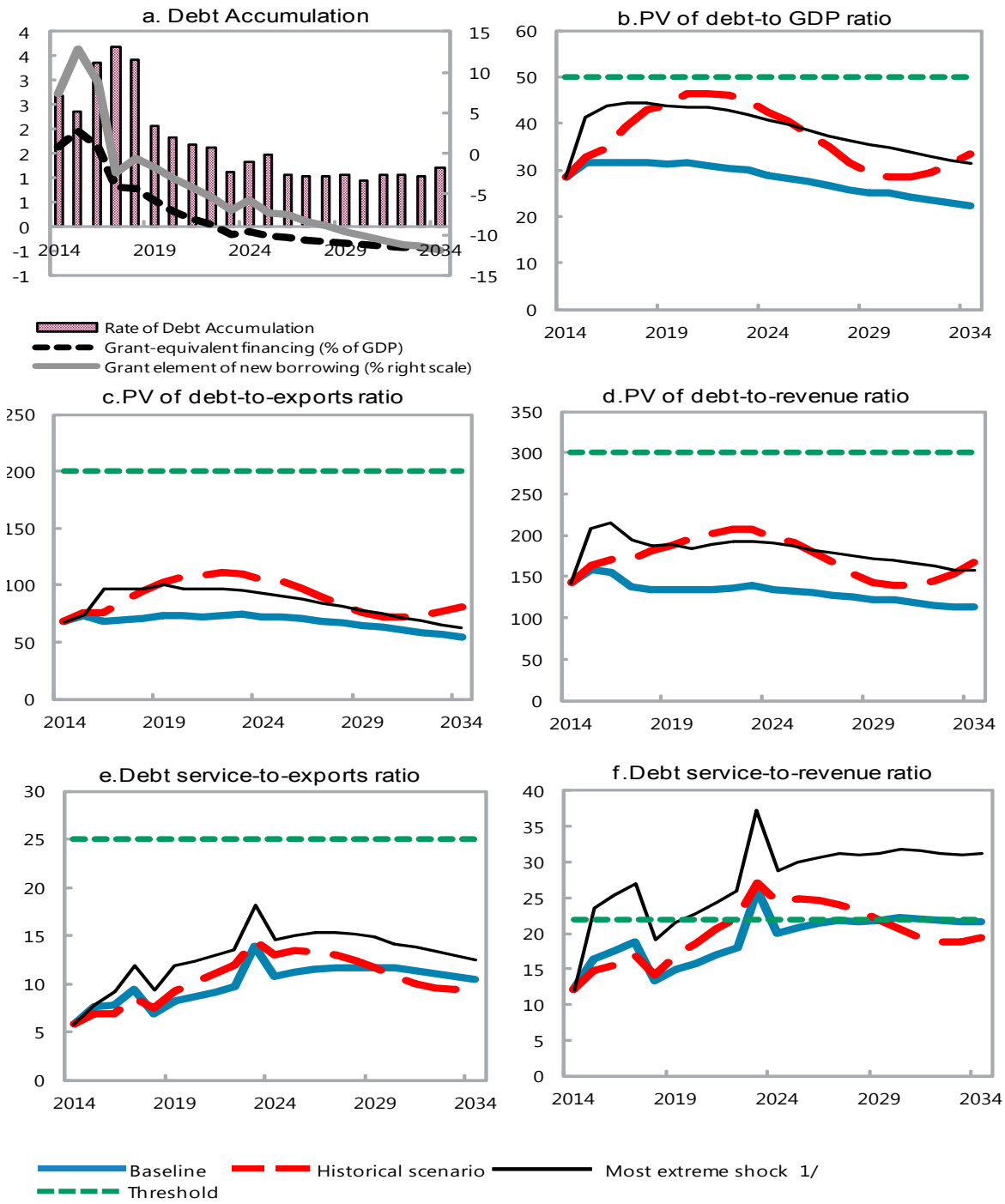
CONCLUSIONS

9. Ghana's public debt situation has worsened since the last DSA. The updated assessment suggests that the risk of distress, while remaining moderate based on an assessment of external public debt, has increased, and Ghana's debt service-to-revenue ratio is approaching high-risk levels. The analysis does not include the liabilities of the central bank—some of which result from the need to sterilize government financing operations—nor the debt of public enterprises that is not guaranteed by the government. Hence, risks could be heightened further, as a result of these institutions' liabilities.

10. Robust growth and fiscal consolidation are essential to maintaining a moderate debt distress rating. Under the baseline, which incorporates the authorities' gradual adjustment plans together with staff's more conservative growth assumptions, the PV of public debt is expected to hover around a still high level of 54 percent of GDP, and the total public debt service-to-revenue ratio is projected to increase above 45 percent in the long run. Additional fiscal adjustment could set these indicators on a more favorable path.

11. The authorities agreed with the thrust of the analysis, but deemed staff's macroeconomic projections too pessimistic. In particular, they stressed that the growth outturn for 2013 is likely to be revised upward and saw stronger economic growth also for 2014.

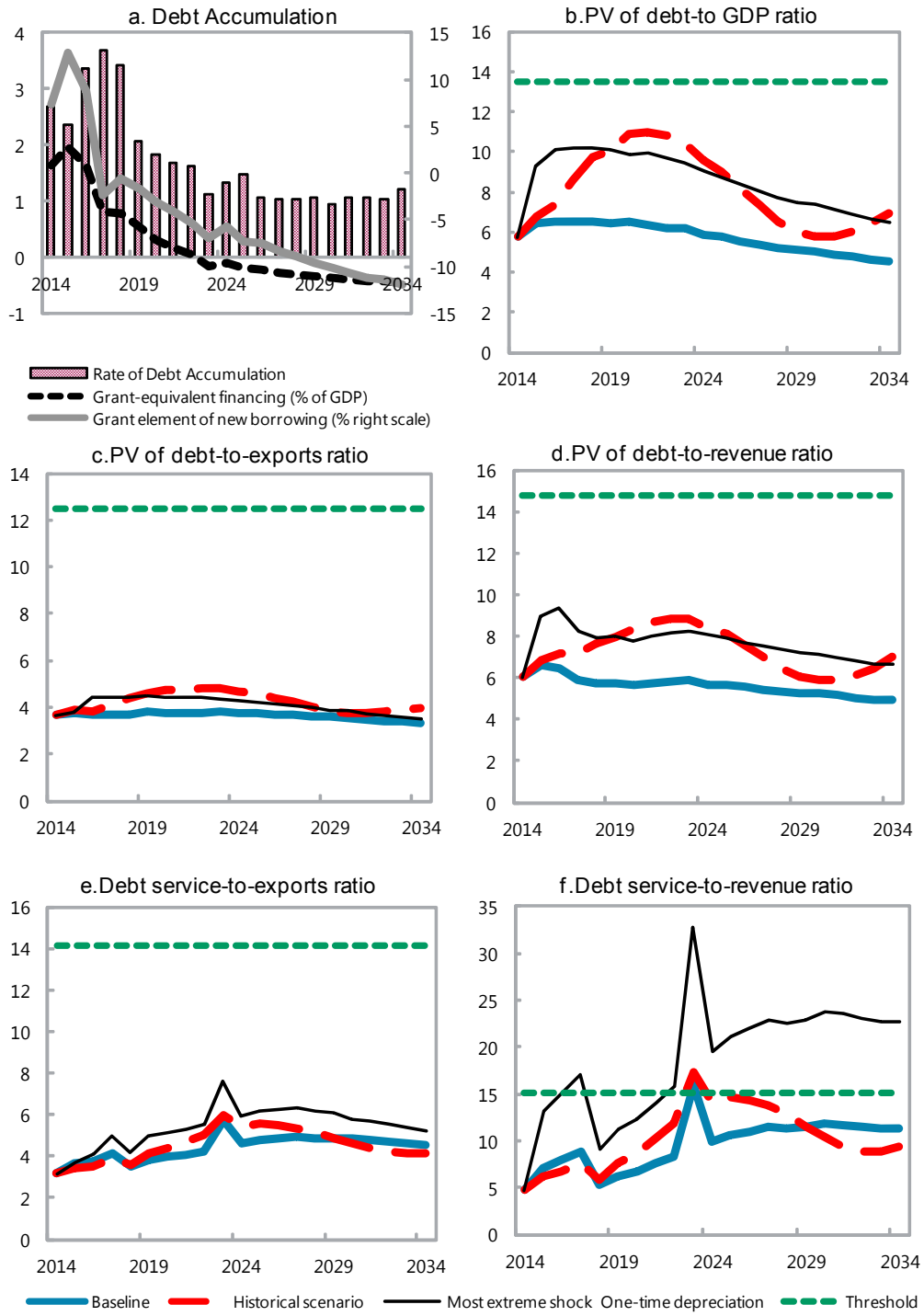
Figure A1. Indicators of Public and Publicly Guaranteed External Debt under Different Scenarios, 2013–2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

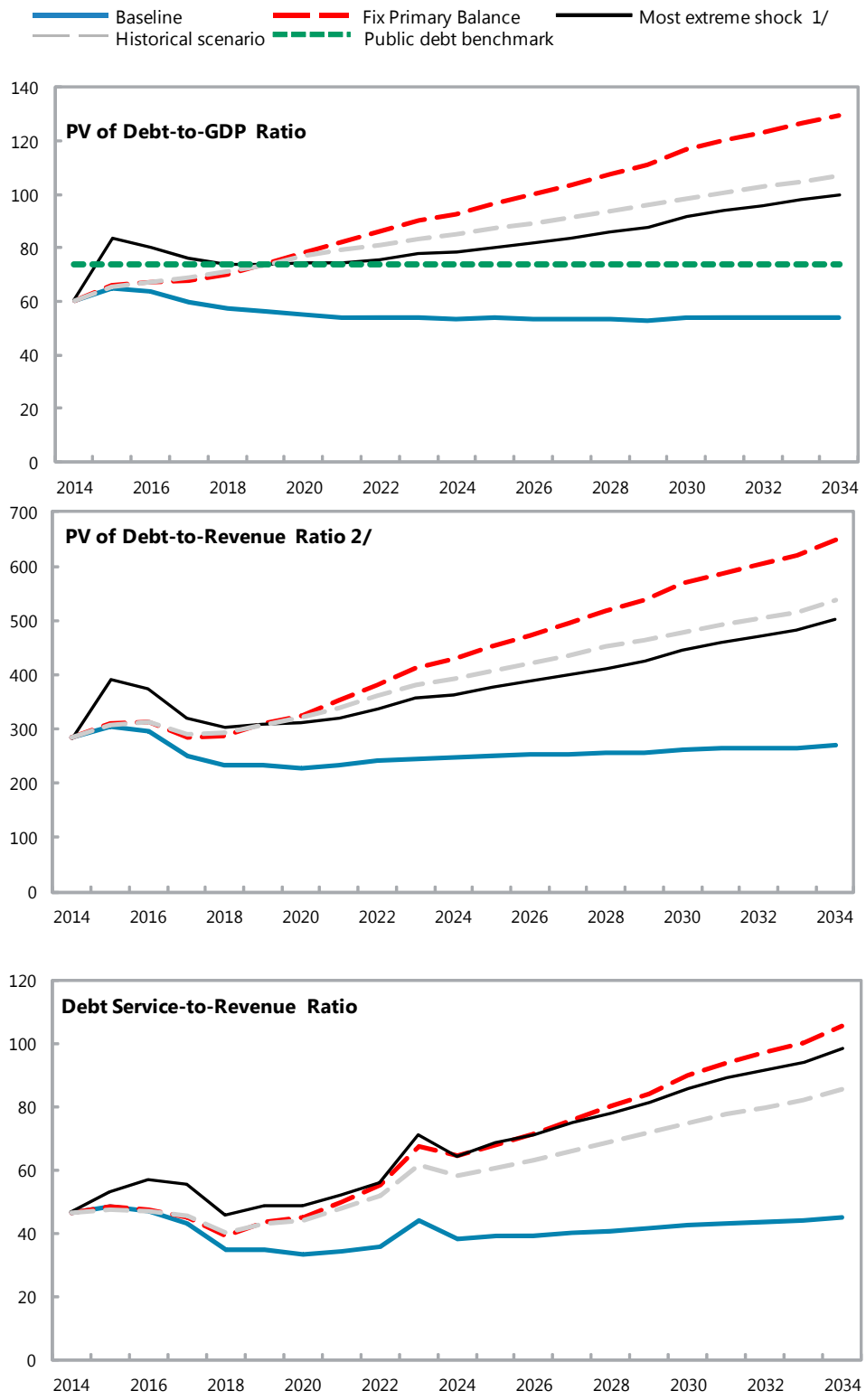
Figure A2. Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Different Scenarios, 2013–2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; and in e. to an Exports shock and in f. to a One-time depreciation shock

Figure A3. Indicators of Public Debt under Different Scenarios, 2013–2033 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table A1. External Debt Sustainability Framework, Baseline Scenario, 2010–2033 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**25.6**	**26.1**	**31.4**			**35.9**	**35.6**	**36.6**	**36.6**	**36.2**	**35.7**		**35.4**	**31.9**	
of which: public and publicly guaranteed (PPG)	19.7	22.6	25.6			32.3	33.8	35.9	35.7	35.4	34.9		32.4	24.5	
Change in external debt	-0.4	0.5	5.3			4.5	-0.4	1.0	0.0	-0.3	-0.6		-0.4	-0.2	
Identified net debt-creating flows	-3.6	3.0	3.7			1.4	-1.6	-2.8	-3.1	-3.0	-1.1		-0.5	-2.5	
Non-interest current account deficit	**8.6**	**11.5**	**12.3**	**8.2**	**2.9**	**9.5**	**6.5**	**5.9**	**5.3**	**5.2**	**5.0**		**5.0**	**1.5**	4.1
Deficit in balance of goods and services	12.7	12.9	14.6			12.3	8.8	8.4	7.5	7.1	6.9		6.6	2.7	
Exports	37.7	41.6	36.3			41.8	42.8	45.9	45.8	45.1	42.7		40.3	41.2	
Imports	50.3	54.4	50.9			54.1	51.6	54.3	53.2	52.3	49.6		46.8	43.9	
Net current transfers (negative = inflow)	-6.7	-5.9	-4.4	-7.7	1.7	-4.6	-4.1	-4.1	-3.7	-3.3	-3.1		-2.1	-0.2	-1.6
of which: official	-0.6	-0.6	-0.2			-0.4	-0.1	-0.1	-0.1	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	2.6	4.6	2.1			1.8	1.8	1.6	1.5	1.4	1.3		0.5	-1.0	
Net FDI (negative = inflow)	**-8.3**	**-8.1**	**-7.3**	**-5.8**	**3.5**	**-7.5**	**-7.5**	**-7.5**	**-7.5**	**-7.5**	**-6.5**		**-5.6**	**-4.0**	-4.9
Endogenous debt dynamics 2/	**-3.8**	**-0.4**	**-1.3**			**-0.6**	**-0.6**	**-1.3**	**-0.9**	**-0.7**	**0.4**		**0.1**	**0.0**	
Contribution from nominal interest rate	0.6	0.7	0.9			1.1	1.3	1.5	1.6	1.5	1.7		1.7	1.6	
Contribution from real GDP growth	-3.2	-1.9	-1.3			-1.7	-1.9	-2.7	-2.5	-2.2	-1.3		-1.6	-1.6	
Contribution from price and exchange rate changes	-1.2	0.9	-0.9			-1.3	-0.7	-0.3	-1.1	-0.6	-0.9		-0.7	...	
Residual (3-4) 3/	**3.2**	**-2.5**	**1.6**			**3.1**	**1.2**	**3.9**	**3.1**	**2.7**	**0.5**		**0.1**	**2.3**	0.4
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	29.7			32.0	33.1	32.1	32.5	32.5	32.0		31.7	29.8	
In percent of exports	...	...	81.7			76.4	77.3	69.9	71.0	71.9	75.0		78.8	72.3	
PV of PPG external debt	**...**	**...**	**23.9**			**28.3**	**31.4**	**31.4**	**31.6**	**31.7**	**31.3**		**28.7**	**22.4**	
In percent of exports	**...**	**...**	**65.9**			**67.8**	**73.3**	**68.5**	**69.2**	**70.1**	**73.4**		**71.3**	**54.4**	
In percent of government revenues	**...**	**...**	**134.8**			**142.4**	**157.6**	**154.3**	**138.3**	**134.1**	**134.7**		**133.6**	**112.5**	
Debt service-to-exports ratio (in percent)	**6.1**	**6.5**	**9.9**			**8.9**	**9.2**	**8.5**	**9.6**	**7.3**	**8.5**		**11.9**	**13.7**	
PPG debt service-to-exports ratio (in percent)	**3.2**	**3.5**	**8.1**			**5.8**	**7.6**	**7.8**	**9.4**	**6.9**	**8.1**		**10.7**	**10.5**	
PPG debt service-to-revenue ratio (in percent)	**6.9**	**8.2**	**16.6**			**12.1**	**16.4**	**17.6**	**18.7**	**13.3**	**14.9**		**20.1**	**21.7**	
Total gross financing need (Millions of U.S. dollars)	987	2447	3784			2224	1196	1013	1040	511	1221		3245	4839	
Non-interest current account deficit that stabilizes debt ratio	9.0	11.0	7.0			5.0	6.8	4.9	5.3	5.5	5.6		5.4	1.7	4.3
Key macroeconomic assumptions															
Real GDP growth (in percent)	15.0	7.9	5.4	7.3	3.0	4.8	5.4	8.1	7.5	6.8	3.8	6.1	4.7	5.4	4.8
GDP deflator in US dollar terms (change in percent)	4.7	-3.3	3.7	7.5	10.3	-15.9	-1.2	-2.5	3.4	3.8	3.8	-1.4	2.5	2.0	2.1
Effective interest rate (percent) 5/	2.7	2.7	3.9	2.5	0.7	3.2	3.7	4.3	4.9	4.6	5.0	4.3	5.1	5.3	5.3
Growth of exports of G&S (US dollar terms, in percent)	54.7	15.1	-4.4	18.4	15.7	1.4	6.7	13.0	10.7	9.3	1.9	7.2	6.1	8.9	6.7
Growth of imports of G&S (US dollar terms, in percent)	39.4	12.8	2.3	19.3	14.9	-6.4	-0.5	10.8	8.9	8.9	2.2	4.0	6.4	6.9	6.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	7.3	12.8	8.8	-2.3	-0.6	-1.6	4.1	-5.8	-11.9	-8.1
Government revenues (excluding grants, in percent of GDP)	17.3	17.7	17.8			19.9	19.9	20.4	22.9	23.6	23.2		21.5	19.9	21.2
Aid flows (in Millions of US dollars) 7/	784.1	641.8	223.5			798.5	864.0	834.5	677.0	644.5	553.0		263.4	115.9	
of which: Grants	784.1	641.8	223.5			514.0	543.3	473.8	451.4	430.2	349.4		105.9	11.4	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.6	2.0	1.6	0.8	0.8	0.6		-0.1	-0.5	-0.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			28.6	31.8	23.1	11.5	14.0	12.2		-2.3	-11.7	-5.4
Memorandum items:															
Nominal GDP (Millions of US dollars)	38752	40436	44223			38944	40561	42756	47528	52661	56749		78101	155146	
Nominal dollar GDP growth	20.4	4.3	9.4			-11.9	4.2	5.4	11.2	10.8	7.8	4.6	7.3	7.6	6.9
PV of PPG external debt (in Millions of US dollars)			9583			10772	11692	13062	14643	16264	17355		22197	33982	
(PVt-PVt-1)/GDPt-1 (in percent)						2.7	2.4	3.4	3.7	3.4	2.1	2.9	1.3	1.2	1.2
Gross workers' remittances (Millions of US dollars)	1942	1760	1524			1341	1330	1402	1402	1424	1442		1344.4	275.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	23.1			27.4	30.4	30.4	30.7	30.8	30.5		28.2	22.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	60.2			62.6	68.1	63.9	65.0	66.1	69.2		68.4	54.1	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	7.4			5.3	7.1	7.3	8.8	6.5	7.7		10.3	10.4	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2. Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	**42.5**	**51.2**	**57.4**			**64.2**	**67.2**	**68.1**	**63.9**	**60.9**	**59.6**		**57.3**	**55.9**	
of which: foreign-currency denominated	19.7	22.6	25.6			32.3	33.8	35.9	35.7	35.4	34.9		32.4	24.5	
Public sector debt, exscluding domestic arrears	**39.5**	**48.0**	**54.3**			**63.7**	**67.2**	**68.1**	**63.9**	**60.9**	**59.6**		**57.3**	**55.9**	
Change in public sector debt	-3.2	8.7	6.2			6.8	3.0	0.9	-4.2	-3.0	-1.3		0.2	-0.1	
Identified debt-creating flows	-3.9	6.5	5.9			6.5	5.2	-1.2	-3.5	-3.2	-1.5		-0.6	-0.1	
Primary deficit	1.3	8.8	5.6	4.1	2.6	3.6	2.7	1.5	-0.8	-0.9	-1.0	0.9	0.6	0.6	0.3
Revenue and grants	19.3	19.3	18.3			21.2	21.3	21.5	23.8	24.4	23.9		21.6	19.9	
of which: grants	2.0	1.6	0.5			1.3	1.3	1.1	0.9	0.8	0.6		0.1	0.0	
Primary (noninterest) expenditure	20.6	28.1	23.9			24.8	24.0	23.0	23.1	23.5	22.9		22.2	20.5	
Automatic debt dynamics	-6.1	-2.2	0.3			2.9	2.6	-2.7	-2.7	-2.3	-0.5		-1.2	-0.8	
Contribution from interest rate/growth differential	-6.1	-2.8	-0.6			-0.1	-0.4	-2.3	-2.1	-1.6	0.1		-0.4	-0.8	
of which: contribution from average real interest rate	-0.2	0.3	2.0			2.5	2.9	2.8	2.6	2.4	2.4		2.2	2.1	
of which: contribution from real GDP growth	-6.0	-3.1	-2.6			-2.6	-3.3	-5.0	-4.7	-4.0	-2.2		-2.6	-2.9	
Contribution from real exchange rate depreciation	0.1	0.6	0.9			3.0	3.0	-0.4	-0.5	-0.6	-0.6		...	...	
Other identified debt-creating flows	0.8	-0.1	0.0			-0.1	-0.1	-0.1	-0.1	-0.1	0.0		0.0	0.0	
Privatization receipts (negative)	1.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.2	-0.1	0.0			-0.1	-0.1	-0.1	-0.1	-0.1	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.8	2.2	0.3			0.4	-2.2	2.1	-0.7	0.3	0.2		0.8	0.0	
Other Sustainability Indicators															
PV of public sector debt	**...**	**...**	**55.7**			**60.3**	**64.8**	**63.6**	**59.8**	**57.2**	**56.0**		**53.6**	**53.7**	
of which: foreign-currency denominated	...	...	23.9			28.3	31.4	31.4	31.6	31.7	31.3		28.7	22.4	
of which: external	...	...	23.9			28.3	31.4	31.4	31.6	31.7	31.3		28.7	22.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	11.3	18.9	20.8			20.2	22.5	21.5	19.1	16.0	15.2		16.3	19.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	305.0			284.1	304.6	296.1	250.9	234.0	234.6		247.7	269.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	313.6			302.9	325.1	312.2	261.3	242.1	240.8		249.3	269.9	
of which: external 3/	...	...	134.8			142.4	157.6	154.3	138.3	134.1	134.7		133.6	112.5	
Debt service-to-revenue and grants ratio (in percent) 4/	25.4	28.1	50.1			46.7	48.3	46.8	43.2	34.8	34.8		38.3	44.8	
Debt service-to-revenue ratio (in percent) 4/	28.4	30.6	51.5			49.8	51.6	49.4	45.0	36.0	35.8		38.5	44.8	
Primary deficit that stabilizes the debt-to-GDP ratio	4.5	0.1	-0.6			-3.2	-0.3	0.7	3.5	2.0	0.3		0.4	0.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	15.0	7.9	5.4	7.3	3.0	4.8	5.4	8.1	7.5	6.8	3.8	6.1	4.7	5.4	4.8
Average nominal interest rate on forex debt (in percent)	3.6	2.9	4.5	1.1	1.8	3.9	4.1	4.5	4.9	4.7	5.1	4.5	5.5	6.8	6.0
Average real interest rate on domestic debt (in percent)	-1.9	0.5	5.2	-0.6	4.3	6.5	7.2	6.4	5.5	5.9	5.5	6.2	5.0	3.4	4.0
Real exchange rate depreciation (in percent, + indicates depreciation	0.4	3.0	3.9	-3.6	7.7	12.0	9.5	-1.2	-1.6	-1.8	-1.7	2.5	-2.4	0.0	0.0
Inflation rate (GDP deflator, in percent)	13.0	13.3	12.3	16.2	5.0	13.1	11.2	9.4	9.4	9.3	9.0	10.2	6.8	6.6	6.8
Growth of real primary spending (deflated by GDP deflator, in percer	0.1	0.5	-0.1	0.1	0.2	0.1	0.0	0.0	0.1	0.1	0.0	0.1	0.1	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	7.3	12.8	8.8	-2.3	-0.6	-1.6	4.1	-5.8	-11.9	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A3. Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–2033

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	28	31	31	32	32	31	**29**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	28	33	35	39	43	44	**42**	33
A2. New public sector loans on less favorable terms in 2014-2034 /2	28	30	33	34	35	35	**37**	37
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	28	29	32	32	32	32	**30**	23
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	28	31	39	39	38	37	**33**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	28	29	31	31	32	31	**29**	22
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	28	33	38	38	38	37	**32**	23
B5. Combination of B1-B4 using one-half standard deviation shocks	28	26	27	27	27	27	**26**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	28	41	44	44	44	44	**41**	31
PV of debt-to-exports ratio								
Baseline	68	73	68	69	70	73	**71**	54
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	68	76	76	86	95	103	**105**	81
A2. New public sector loans on less favorable terms in 2014-2034 /2	68	71	73	75	78	83	**91**	89
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	68	67	67	67	68	72	**71**	53
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	68	74	97	97	97	100	**92**	63
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	68	67	67	67	68	72	**71**	53
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	68	76	83	83	83	86	**80**	55
B5. Combination of B1-B4 using one-half standard deviation shocks	68	62	61	62	63	67	**67**	52
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	68	67	67	67	68	72	**71**	53
PV of debt-to-revenue ratio								
Baseline	142	158	154	138	134	135	**134**	112
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	142	164	170	171	182	188	**197**	168
A2. New public sector loans on less favorable terms in 2014-2034 /2	142	152	164	150	149	152	**170**	184
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	142	146	157	141	137	138	**139**	115
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	142	154	191	170	163	161	**151**	114
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	142	147	153	138	134	134	**135**	112
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	142	164	187	166	159	158	**149**	113
B5. Combination of B1-B4 using one-half standard deviation shocks	142	132	131	118	115	117	**120**	102
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	142	208	215	193	188	189	**190**	158

Table A3. Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–2033 (continued)

Debt service-to-exports ratio								
Baseline	6	8	8	9	7	8	**11**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	6	7	7	8	7	9	**13**	9
A2. New public sector loans on less favorable terms in 2014-2034 /2	6	8	7	8	6	7	**11**	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	6	8	8	9	7	8	**11**	10
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	6	8	9	12	9	12	**15**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	6	8	8	9	7	8	**11**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	6	8	8	10	8	10	**13**	11
B5. Combination of B1-B4 using one-half standard deviation shocks	6	7	8	9	6	7	**10**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	6	8	8	9	7	8	**11**	10
Debt service-to-revenue ratio								
Baseline	12	16	18	19	13	15	**20**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	12	15	16	17	14	17	**24**	19
A2. New public sector loans on less favorable terms in 2014-2034 /2	12	16	16	17	11	13	**20**	34
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	12	17	18	20	14	16	**21**	23
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	12	16	18	21	16	19	**24**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	12	17	18	19	14	15	**21**	22
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	12	16	19	21	16	19	**23**	22
B5. Combination of B1-B4 using one-half standard deviation shocks	12	16	16	17	12	13	**18**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	12	23	25	27	19	21	**29**	31
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	-14	-14	-14	-14	-14	-14	**-14**	-14

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A4. Sensitivity Analysis for Key Indicators of Public Debt, 2013–2033

	Projections								
	2014	2015	2016	2017	2018	2019	2023	2024	2034
PV of Debt-to-GDP Ratio									
Baseline	60	65	64	60	57	56	54	54	54
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	60	65	67	69	71	73	84	85	107
A2. Primary balance is unchanged from 2014	60	66	67	68	70	74	90	93	129
A3. Permanently lower GDP growth 1/	60	65	65	62	60	60	65	66	95
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	60	66	68	65	64	64	67	68	80
B2. Primary balance is at historical average minus one standard deviations in 2015-201	60	69	74	70	67	66	64	64	64
B3. Combination of B1-B2 using one half standard deviation shocks	60	68	72	69	66	66	66	66	71
B4. One-time 30 percent real depreciation in 2015	60	83	80	76	74	74	78	78	100
B5. 10 percent of GDP increase in other debt-creating flows in 2015	60	76	74	70	68	66	65	64	65
PV of Debt-to-Revenue Ratio 2/									
Baseline	284	305	296	251	234	235	246	248	270
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	284	307	314	289	292	308	382	392	537
A2. Primary balance is unchanged from 2014	284	309	312	285	287	310	412	429	650
A3. Permanently lower GDP growth 1/	284	307	302	260	247	253	296	307	479
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	284	309	317	274	261	268	306	313	399
B2. Primary balance is at historical average minus one standard deviations in 2015-201	284	325	343	292	274	275	292	294	322
B3. Combination of B1-B2 using one half standard deviation shocks	284	318	335	287	271	275	300	304	355
B4. One-time 30 percent real depreciation in 2015	284	392	374	320	303	308	356	362	502
B5. 10 percent of GDP increase in other debt-creating flows in 2015	284	356	346	296	277	278	296	298	326
Debt Service-to-Revenue Ratio 2/									
Baseline	47	48	47	43	35	35	44	38	45
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	47	48	47	45	40	43	61	58	86
A2. Primary balance is unchanged from 2014	47	48	47	45	39	44	67	64	105
A3. Permanently lower GDP growth 1/	47	49	47	44	36	37	51	46	76
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	47	49	49	46	38	39	53	47	65
B2. Primary balance is at historical average minus one standard deviations in 2015-201	47	48	49	50	43	41	51	46	53
B3. Combination of B1-B2 using one half standard deviation shocks	47	48	49	49	41	40	52	47	59
B4. One-time 30 percent real depreciation in 2015	47	53	57	55	46	48	71	64	98
B5. 10 percent of GDP increase in other debt-creating flows in 2015	47	48	51	54	40	42	52	46	54

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.